

Ten one-liners that are quietly wrong

Each one runs without an error and prints a plausible number.

The number is not what you think it is.

Your coverage number is too high

X WHAT PEOPLE WRITE

```
samtools depth in.bam | awk '{s+=$3} END  
{print s/NR}'
```

✓ WHAT IT SHOULD BE

```
samtools depth -a in.bam | awk '{s+=$3} END  
{print s/NR}'
```

Without -a, samtools depth omits zero-coverage positions entirely, so dividing by NR gives mean depth over covered bases only.

You are counting the same read twice

X WHAT PEOPLE WRITE

```
samtools view -c -F 4 in.bam
```

✓ WHAT IT SHOULD BE

```
samtools view -c -F 0x904 in.bam
```

-F 4 also counts secondary (0x100) and supplementary (0x800) alignments, so it over-reports.

Your unmapped reads lost their pairing

X WHAT PEOPLE WRITE

```
samtools fastq -f 4 in.bam > unmapped.fq
```

✓ WHAT IT SHOULD BE

```
samtools fastq -f 4 -1 un_R1.fq -2 un_R2.fq  
-0 /dev/null -s /dev/null -n in.bam
```

`samtools fastq -f 4 in.bam > out.fq` interleaves R1 and R2 into one file and drops singletons silently.

Your gene names are silently wrong

X WHAT PEOPLE WRITE

```
awk -F'\t' '$3=="gene" {print $9}' in.gtf |  
awk -F';' '{print $1, $3}'
```

✓ WHAT IT SHOULD BE

```
awk -F'\t' '$3=="gene" && match($9,  
/gene_name "[^"]+"/) {print substr($9,  
RSTART+11, RLENGTH-12)}' in.gtf
```

Positional parsing (`cut -d';' -f3`) breaks on genes with no `gene_name` - common for novel and lncRNA genes - and silently returns `gene_source` instead.

The pipeline failed and said nothing

X WHAT PEOPLE WRITE

```
samtools view in.bam | wc -l
```

✓ WHAT IT SHOULD BE

```
set -o pipefail; samtools view in.bam | wc  
-l
```

By default a pipeline reports only the LAST command's status, so a crashed samtools upstream of wc -l looks like success.

gzip is not bgzip

X WHAT PEOPLE WRITE

```
gzip -c in.vcf > in.vcf.gz && tabix -p vcf  
in.vcf.gz
```

✓ WHAT IT SHOULD BE

```
bgzip -c in.vcf > in.vcf.gz && tabix -p vcf  
in.vcf.gz
```

A plain gzip file cannot be indexed.

merge trusted you to sort first

X WHAT PEOPLE WRITE

```
bedtools merge -i regions.bed
```

✓ WHAT IT SHOULD BE

```
sort -k1,1 -k2,2n regions.bed | bedtools  
merge -i -
```

bedtools merge assumes sorted input.

Your subsampled mates no longer match

X WHAT PEOPLE WRITE

```
seqtk sample -s100 R1.fq.gz 10000 > s1.fq;  
seqtk sample -s42 R2.fq.gz 10000 > s2.fq
```

✓ WHAT IT SHOULD BE

```
seqtk sample -s100 R1.fq.gz 10000 > s1.fq;  
seqtk sample -s100 R2.fq.gz 10000 > s2.fq
```

Using a different -s seed for R1 and R2 silently destroys pairing.

pigz did not speed that up

X WHAT PEOPLE WRITE

```
pigz -dc big.fq.gz | wc -l
```

✓ WHAT IT SHOULD BE

```
gzip -dc big.fq.gz | wc -l    # same speed;  
use pigz -p N to COMPRESS
```

pigz parallelises compression only.

LC_ALL=C is fast, and dangerous

X WHAT PEOPLE WRITE

```
sort -k1,1 -k2,2n big.bed | join - ids.txt
```

✓ WHAT IT SHOULD BE

```
LC_ALL=C sort -k1,1 -k2,2n big.bed |  
LC_ALL=C join - ids.txt
```

LC_ALL=C is a large speed-up but it changes collation order.

All of it, tested.

- 66 command-line recipes
- 43 documented traps
- A 300 KB toy genome, committed
- Every command run in CI on each push

Clone it and every command in the README runs immediately.
Nothing to download.

github.com/GeneticistHere/bioinformatics-bash-cookbook